

NOT ALL CHANGES MATTER: DECISION-ALIGNED DELTA SELECTION FOR SPEECH RECOGNITION

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ABSTRACT

Recent studies improve speech recognition adaptation by incorporating pretrained-to-fine-tuned representation shifts with other features, but treat these changes as a whole. Whether adapted recognizers rely on all shift coordinates remains underexplored. We propose **Decision-Aligned Delta Selection (DADS)**, a task-aware framework that identifies decision-supporting representation shifts by aligning feature changes with the adapted recognizer's decision sensitivity. We find that adaptation gains are concentrated in a subset of fine-tuning-induced representation shifts in the encoder's native output basis, rather than uniformly distributed across coordinates. A released L2-ARCTIC recognizer maintains nearly full accuracy after discarding 25% of shifts (10.58% vs. 10.50% WER), and DADS preserves substantially more performance than magnitude selection when retaining only half of the shifts (11.71% vs. 16.64%). Further removal and rescaling analyses demonstrate that the contribution of a shift depends on both its decision relevance and retained magnitude.

Index Terms— speech recognition, self-supervised learning, model adaptation, representation analysis, feature selection

1. INTRODUCTION

Modern speech recognition systems increasingly rely on large-scale self-supervised speech representations, which provide strong initialization for adapting models to new speakers, accents, and domains [1, 2, 3, 4, 5, 6]. Through supervised fine-tuning, these models acquire task-specific capabilities from limited labeled speech and achieve substantial improvements under distribution shifts [7, 8, 9]. Adaptation reshapes the learned representation space by introducing substantial differences between pretrained and adapted representations, yet how such information is organized within the representation space remains poorly understood.

Recent studies have demonstrated that differences between pretrained and adapted models contain transferable information. Task arithmetic transfers model behaviors through parameter differences, while TIES-Merging and DARE selectively manipulate such differences for model combination [10, 11, 12]. Similar ideas have also been explored in automatic speech recognition through task vector algebra and selective model merging [13, 14]. In representation space, *Mind the Shift* exploits delta embeddings, defined as the difference between pretrained and fine-tuned speech representations, to enhance child ASR; accent steering further utilizes representation differences between speech groups to control adaptation behavior [15, 16].

Although existing methods vary in how representation differences are constructed and utilized, they generally treat representation shifts as a unified adaptation signal without considering how individual changes affect the adapted recognizer. Our key insight

is that shift relevance changes across the representation-to-decision boundary: from feature movement in the representation space to decision sensitivity in the adapted recognizer. A large shift may have limited impact if the recognizer is insensitive to that direction, while a smaller shift can be critical when aligned with a decision-sensitive direction.

Two observations support this decision-aligned view. First, fine-tuning-induced shifts exhibit highly non-uniform utility across native output coordinates. Under a fixed retention budget, preserving selected shifts can maintain adaptation performance, while removing different groups of the same size leads to substantially different recognition outcomes. Second, shift magnitude alone cannot explain adaptation utility. Useful shifts emerge from the interaction between representation movement and the recognizer's decision sensitivity, rather than from either factor independently.

Based on these observations, we propose **Decision-Aligned Delta Selection (DADS)**, a decision-aligned pruning framework for fine-tuning-induced representation shifts. DADS first decomposes adaptation into native output coordinates and evaluates each coordinate by measuring how its change affects the adapted CTC decision. Specifically, DADS combines shift magnitude with CTC objective sensitivity to identify decision-supporting shifts, then selectively retains informative changes while removing redundant adaptation signals. The intervention keeps the adapted CTC head and decoding fixed, so retention and reversion directly test which changes are sufficient and necessary for the existing recognizer (Fig. 1).

We evaluate DADS on L2-ARCTIC across adapted ASR models [17] and reveal substantial redundancy in fine-tuning-induced shifts. DADS retains 98.8% of the full adaptation gain using only 50% of native output coordinates, achieving 11.71% WER compared with 16.64% for magnitude-based selection. Counterfactual reversion further shows that high-utility shift groups are more decision-relevant, causing substantially larger degradation when removed than low-utility groups.

Our main contributions are summarized as follows:

1. We identify that adaptation utility is concentrated in a decision-dependent subset of native-coordinate fine-tuning shifts, showing why shift magnitude alone is insufficient.
2. We align representation displacement with adapted CTC sensitivity to select decision-supporting native-coordinate shifts and remove redundant changes without retraining.
3. We use frozen-head retention and reversion to establish which shifts the adapted recognizer functionally relies on and to verify the value of decision-aligned selection.

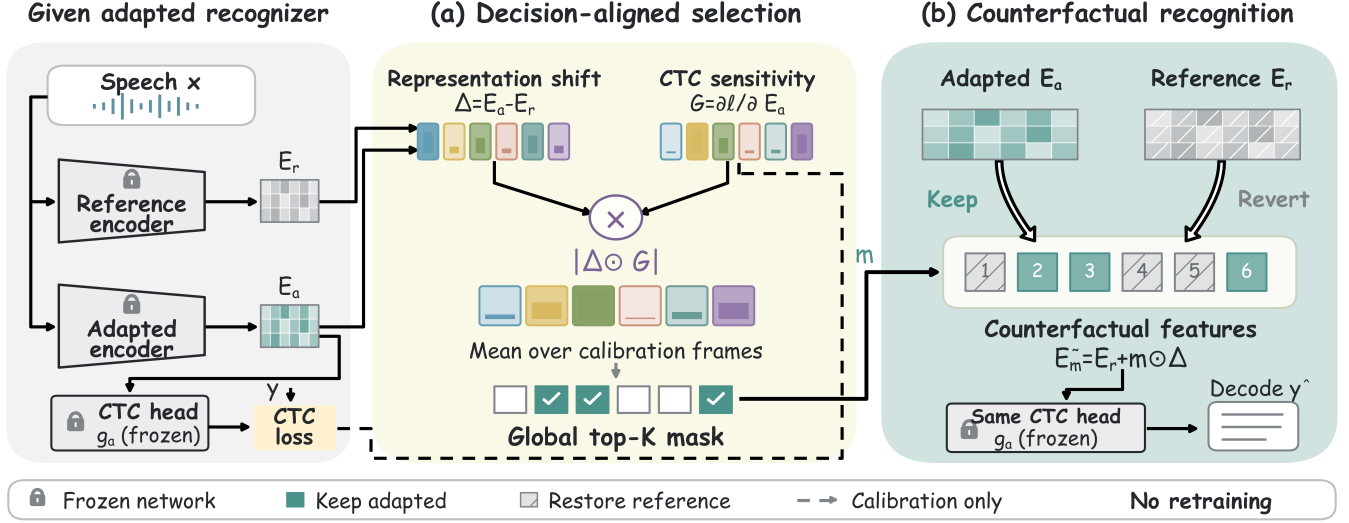


Fig. 1. Overview of DADS. (a) Shift Δ is combined with CTC sensitivity $G = \partial \ell / \partial E_a$ from the frozen adapted head. Averaging absolute elementwise products over valid training frames gives a global top- K native-coordinate mask. (b) Selected coordinates keep adapted values; others restore reference values. The same frozen head decodes without retraining. Labels and gradients are calibration-only; feature strips schematically show corresponding coordinates.

2. DECISION-ALIGNED DELTA SELECTION

2.1. Keeping some changes and undoing others

As shown in Fig. 1, the reference and adapted encoders map utterance x to final-layer features $E_r(x), E_a(x) \in \mathbb{R}^{T_x \times d}$, where T_x is the number of output frames and d is the number of native output coordinates. Their difference is

$$\Delta(x) = E_a(x) - E_r(x). \quad (1)$$

Both encoders receive the same waveform and have the same frame resolution. When the reference is the recorded initialization, Δ is the change induced by fine-tuning. We also apply the operation to a released recognizer and a separately obtained reference, as specified in Section 3.

A binary mask $m \in \{0, 1\}^d$ retains K native output coordinates of the change:

$$\tilde{E}_m(x) = E_r(x) + m \odot \Delta(x). \quad (2)$$

The mask is broadcast over frames. For example, retaining 50% uses adapted values in 512 of the 1024 native coordinates and reference values in the other 512. The input to the recognition head still has 1024 coordinates. Keeping every change gives the complete adapted representation (*Full*); keeping none gives the reference representation (*NoShift*). A single mask is shared across all utterances.

Every condition uses the adapted affine CTC head $g_a(h) = W_a h + b_a$ and greedy decoding. NoShift therefore sends reference features through this same head. This comparison asks what the *adapted recognizer* needs from its feature changes. Because neither the encoder nor the head is retrained after masking, a WER difference directly records the effect of substituting the selected features. All masks are defined in the native final-layer coordinate system exposed to W_a , which is kept fixed across models, budgets, and interventions.

At each frame, the mask changes the logits by

$$g_a(\tilde{E}_{m,t}) - g_a(E_{a,t}) = W_a((m - 1) \odot \Delta_t). \quad (3)$$

Thus, the effect of a feature change depends on how the recognition head uses its coordinate. Its magnitude alone does not account for that dependence. This motivates scoring changes with the recognition loss.

2.2. Scoring changes with the CTC loss

On a labeled subset $\mathcal{D}_u$ of the adaptation training data, we compute the loss at the complete adapted features:

$$\ell_{x,y} = \frac{1}{|y|} \mathcal{L}_{\text{CTC}}(g_a(E_a(x)), y), \quad (4)$$

where $|y|$ is the transcript length in characters. CTC sums over valid frame-to-label alignments [18]; no forced alignment is required for this score. The length normalization gives each utterance a loss per target character.

Undoing a change in native coordinate i perturbs each frame by $-\Delta_{t,i}$. In a first-order approximation, its effect on the loss involves the product of the change and the corresponding gradient. Following Taylor saliency [19], DADS aggregates the absolute products:

$$U_i = \frac{1}{N_u} \sum_{(x,y) \in \mathcal{D}_u} \sum_{t=1}^{T_x} \left| \Delta_{t,i}(x) \frac{\partial \ell_{x,y}}{\partial E_{a,t,i}} \right|, \quad (5)$$

where $N_u = \sum_{(x,y) \in \mathcal{D}_u} T_x$. We take the absolute value at each frame before averaging, so positive and negative products do not cancel. Only valid frames contribute. The result is one score per native output coordinate, computed from a single pass through cached features and the fixed head.

DADS keeps the K native coordinates with the largest U_i . The score combines two requirements: a coordinate must have changed,

and the recognition loss must be sensitive to that change. A small gradient reduces the score even when the change is large; a small change reduces it even when the loss is sensitive to that coordinate. The subsequent masking experiments measure the actual recognition effect of keeping or undoing groups selected by this local score.

For comparison, *Magnitude* replaces the product in Eq. (5) with $|\Delta_{t,i}|$, while *Gradient* uses $|\partial \ell_{x,y} / \partial E_{a,t,i}|$. Both use the same valid-frame average and retain the same number of coordinates. *Random* selects exactly K coordinates uniformly. We also undo either the highest- or lowest-scoring quarter. Undoing the lowest quarter is the same mask as retaining the highest 75%; undoing the highest quarter tests the opposite end of the ranking.

2.3. Rescaling retained changes

A mask determines where the representation changes, but also reduces the overall difference from the reference. Following the drop-and-rescale idea in DARE [12], we compare Random with

$$\tilde{E}_m^{\text{scaled}} = E_r + \frac{d}{K} m \odot \Delta. \quad (6)$$

The retained changes are doubled at 50% retention and multiplied by 4/3 at 75%. For a uniformly sampled mask of exactly K coordinates, each coordinate is retained with probability K/d , so the scaled difference equals Δ in expectation over masks. The same random masks are used with and without scaling.

This control changes the amplitude while keeping the selected coordinates fixed. On D0, we also apply the two scales, 1 and d/K , to the DADS and Magnitude rankings. Both are reported at every budget, with neither chosen from development or test results. This separates the effect of rescaling a given selection from the effect of changing its ranking.

3. EXPERIMENTAL SETUP

3.1. Data and adapted models

We use L2-ARCTIC [17] and the splits released by Thai et al. [7]. Each split uses 18 speakers for training and development and six held-out speakers for testing, across the same six first-language groups. Folds 0–2 have 16,312, 16,070, and 15,953 training utterances, respectively. Development sets contain 1,867, 1,979, and 2,086 utterances; test sets contain 675, 686, and 691. The evaluation therefore measures transfer to held-out speakers within these accent groups.

R0 is the released wav2vec 2.0 Large recognizer adapted to Fold 0 with CTC and supervised contrastive learning [20, 7]. Its reference is the LibriSpeech-960h wav2vec 2.0 Large checkpoint [21]. The released model and reference differ in feature-extractor normalization. R0 evaluates selection between these two checkpoints in their native final-layer coordinates.

W1/W2 adapt wav2vec 2.0 Large on Folds 1/2 with CTC and supervised contrastive learning; D0 adapts data2vec Audio Large on Fold 0 with CTC. For these three models, the reference is the recorded initialization, so the differences measure fine-tuning changes. Both source models have 960 h of LibriSpeech ASR supervision. All features have $d = 1024$, and every adapted head has 32 output symbols.

The local runs warm up the head for one epoch at learning rate 3×10^{-6} , then jointly train at 10^{-5} with frozen convolutional features. Development WER selects the checkpoint within 40 epochs. W1/W2 use two GPUs, batch size 24 per GPU, a linear schedule,

Table 1. Test WER (%) after retaining selected changes. R0 uses a separate reference checkpoint; W1/W2/D0 use their fine-tuning initialization. Full and NoShift retain all and none of the changes, respectively.

Selection	R0	W1	W2	D0
Full	10.50	15.14	14.46	11.39
DADS, 75%	10.58	15.36	14.54	10.96
DADS, 50%	11.71	16.01	15.58	12.94
Magnitude, 50%	16.64	16.63	16.48	13.75
NoShift	118.91	20.33	19.15	13.77

contrastive weight 0.05, temperature 0.1, and projection dimension 256. D0 uses batch size four on one GPU and a cosine schedule. Each model comes from one adaptation run with seed 1337. Comparisons between selection methods use the same adapted model, reference, head, and retained coordinate count.

3.2. Calibration and evaluation

We select every tenth training utterance after sorting identifiers within each speaker, giving 1,640/1,615/1,600 calibration utterances for Folds 0/1/2. These utterances are part of the adaptation training data; their labels determine the ranking. Development and test utterances are not used to score coordinates. Final-layer features are cached in float32 from the same 16-kHz waveforms. After computing a ranking, evaluation only composes the selected features and applies the fixed head.

All conditions use complete utterances and greedy CTC decoding without a language model. WER is the total word edit distance divided by the total reference word count. We report retention budgets of 25%, 50%, and 75% where available, and compare high- and low-score removal at 25%. Random results report mean and sample SD over mask seeds 1337, 2027, and 31415. These are different masks of the same model. R0 intervals use 10,000 paired utterance-bootstrap samples from the existing predictions.

3.3. Computation

All selection methods use the same two encoders. On an RTX 3090 with FP32 and batch size one, this takes 42.25–47.79 ms per utterance across the tested models and masks, versus 20.74–29.06 ms for the adapted encoder alone. The measurements use eight test utterances per model, five timed repeats, and exclude audio I/O and waveform preprocessing. The mask changes the features sent to the head; it leaves both encoders’ computation intact. Caching allows the same encoded utterances to be reused across all selection controls.

4. RESULTS AND ANALYSIS

4.1. Recognition with selected changes

On R0, DADS retains 75% of the changes at 10.58% WER, close to 10.50% for Full (Table 1). The difference is 0.09 percentage points, with a paired 95% interval of $[-0.21, 0.38]$. At 50%, DADS gives 11.71%, compared with 16.64% for Magnitude: a 29.6% relative WER reduction. The paired difference is -4.92 points, with interval $[-5.72, -4.16]$.

At 50%, Gradient and Random give 50.43% and $46.17 \pm 26.91\%$ WER; Random+Rescale gives $35.87 \pm 25.42\%$. DADS at 11.71%

Table 2. R0 utility mass contained in the top native coordinates. Rotated values are mean $\pm$ SD over five equivalent orthogonal reparameterizations.

Top coordinates	Native	Orthogonal rotations
1%	7.78	3.37 ± 0.22
5%	20.23	12.68 ± 0.39
10%	30.30	21.55 ± 0.37
25%	50.93	42.38 ± 0.18

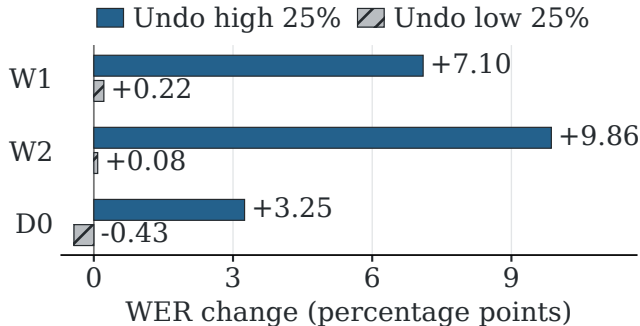


Fig. 2. WER change when the highest- or lowest-scoring quarter of fine-tuning changes is undone. Both conditions restore 256 native coordinates to the reference values and use the same adapted head. Positive values mean worse recognition.

outperforms both score components, and rescaling random changes does not close the gap.

At 50%, DADS improves over Magnitude by 0.62/0.91/0.80 points for W1/W2/D0. At 75%, W1/W2 are within 0.22/0.08 points of Full and D0 reaches 10.96%; Full beats NoShift throughout. Thus, much of the adaptation change can be restored with little loss.

4.2. Native-basis concentration control

DADS ranks changes at the final encoder-head interface. For orthogonal Q , we use the equivalent parameterization $E' = QE$, $W'_a = W_a Q^\top$, and unchanged b_a , so $W'_a E' + b_a = W_a E + b_a$. On R0, five random bases preserve decoded predictions and WER (8.590% dev; 10.499% test). Native utility mass exceeds the rotated null at every top- K level (Table 2); top-1% Jaccard overlap is zero and top-10% overlap is 0.058 ± 0.021 . The concentration therefore reflects alignment with the fixed native output interface used by DADS.

4.3. Uneven reliance on native-coordinate changes

A small WER change after undoing the lowest-scoring quarter could reflect tolerance to removing many coordinates. With a fixed quarter reversion (Fig. 2), undoing the highest-scoring coordinates raises WER by 7.10/9.86/3.25 points for W1/W2/D0, whereas undoing the lowest changes it by 0.22/0.08/−0.43 points.

The interventions share the model, head, utterances, and coordinate count; their contrast shows non-uniform reliance on native-coordinate changes, with the training-derived ranking transferring to held-out speakers.

Table 3. W1/W2 test WER (%) at 50% retention. Random results are mean $\pm$ SD over three masks, reused for rescaling. Bold marks the lowest mean WER in each column.

Method	W1	W2
DADS	16.01	15.58
Magnitude	16.63	16.48
Gradient	19.79	20.48
Random	23.39 ± 0.58	24.20 ± 0.81
Random+Rescale	15.64 ± 0.13	14.93 ± 0.08

Table 4. D0 test WER (%) with the same two prespecified scales applied to each ranking. Bold marks the lowest WER at each budget.

Retained	DADS		Magnitude	
	1	d/K	1	d/K
25%	15.20	18.81	13.30	12.72
50%	12.94	13.54	13.75	12.00
75%	10.96	12.07	11.10	11.59

4.4. Interaction between selection and scale

At 50%, DADS beats Magnitude, Gradient, and Random in W1/W2 (Table 3). Rescaling the same random masks reduces WER from 23.39/24.20% to 15.64/14.93%, below DADS at 16.01/15.58%; at 75%, all methods are close (15.31–15.36% for W1; 14.54–14.62% for W2).

D0 likewise improves from $15.08 \pm 0.31\%$ to $11.95 \pm 0.33\%$ with random rescaling at 50%; Table 4 applies both scales to fixed DADS/Magnitude coordinates.

At 50%, scaling Magnitude improves 13.75% to 12.00%, but scaling DADS worsens 12.94% to 13.54%; at 25%, the effects are 13.30% to 12.72% and 15.20% to 18.81%. At 75%, neither benefits and unscaled DADS is best (10.96%). Scale therefore interacts with ranking and budget.

4.5. Sensitivity to calibration size

With the model and head fixed, 128 calibration utterances keep DADS within 0.2 WER points of the full-set results at both 50% and 75%. Top-quarter overlap with the full ranking rises from 81.9% at 128 utterances to 91.3% at 512 and 94.9% at 1,024, while WER remains similar; a small sample is sufficient for useful selection.

5. CONCLUSION

DADS identifies decision-aligned native-coordinate shifts: CTC sensitivity selects changes that preserve recognition, while reversion confirms their unequal reliance. The equivalent-basis control ties this concentration to the fixed native output interface; scale effects depend on the retained coordinates.

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